

Test Coverage Analysis — 2026-06-16

SECTION 06-operations-it

TYPE document

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Scope note: **The-Office is a documentation/spec repository, not a code repository.** There is no source to instrument, so this is not a line/branch-coverage report. It is a **spec-vs-test gap audit**: it cross-references the testable behaviour described in the `SPEC-*.md` files, `captadel-plan.md`, and `test-ready.md` against what those same documents claim is covered, and flags the gaps. The implementation and its real test suites live in separate repos (`captadel/evals/**`, `sales_agents/tests/**`, `functions/**`).

1. What is being tested today

Two distinct test surfaces are described in this repo:

Surface	Where described	Claimed coverage	Runner
Sales-agent E2E (Enrollment / CFI / B2B)	<code>test-ready.md</code>	55 cases (≥ 38 min)	<code>python3</code> <code>sales_agents/tests/test_runner.py</code>
Captain Adel RAG eval harness	<code>spec-captain-adel-refusal-protocol.md</code> §7, <code>captadel-plan.md</code>	target ≥ 95 cases (v0.5); ≥ 80 (v0.4)	<code>evals/run.js</code> + CI <code>.github/workflows/eval.yml</code>

Three of the five specs define little or no test plan at all (see §3).

2. Gaps the documents themselves already flag (highest priority)

These are not inferred — the specs explicitly mark them **uncovered**. They should be the first additions because the authors already decided they belong in the suite.

- Four refusal cases marked "currently uncovered"** (`refusal-protocol` §7.2) — ready-to-paste JSON fixtures already exist in the spec:
 - `refuse-personal-safety` (fatigue / fitness-to-fly, class 3.2)
 - `refuse-enable-noncompliance` (below-minima "what's the risk", class 3.3)
 - `ar-refuse-unverifiable-limit` (Arabic unverifiable-limit refusal, class 1.1)
 - `refuse-conflicting-sources` (two corpus passages, different versions)
- Tool-use eval cases do not exist yet** (`captadel-plan.md` Evaluation notes). E6B / W&B / fuel / VFR-minima compute tools introduce a new failure class — *confidently wrong calculation* —

and the plan states no compute tool should ship before cases assert: right tool called, right inputs, working shown, **rule still cited**. Currently zero coverage.

3. **Multi-tenant prompt isolation** has no eval case (`captadel-plan.md`). As tenants beyond Fly GACA land, there is no test that one tenant's system prompt / persona cannot leak into another's response. Needs an explicit isolation contract + red-team case.
4. **Runtime injection logging** → **eval growth loop** is described but not in place: the adversarial eval set is static; there is no runtime detector feeding real injection attempts back into the suite.

3. Specs with weak or absent acceptance criteria

Spec	Lines	Test/acceptance language	Risk
<code>spec-crm.md</code>	196	None	High
<code>spec-freshness-pipeline.md</code>	108	None	High
<code>spec-currency-tracker.md</code>	788	Regulatory [VERIFY] TODOs only; one "logbook entry validation" note; no automated test plan	Critical
<code>spec-instructor-dashboard.md</code>	243	Good — mandates Firebase-emulator rule tests before deploy	Low
<code>spec-captain-adel-refusal-protocol.md</code>	1025	Strong — full §7 eval harness spec	Low (but see §2)

3.1 Currency tracker — the critical gap

This computes **legal pilot-currency status** from GACAR Part 61 rules, yet defines no test cases for any calculation:

- 90-day passenger-currency threshold (day vs. night, category/class).
- Night-currency window (1 hr after sunset → 1 hr before sunrise).
- IPC trigger at > 6 calendar months instrument-lapse.
- Flight-review 24-month window and the § 61.21(d) bypass list.
- The combination paths (iv)(v)(vi) the spec itself calls "complex — confirm all."

These are date-boundary, legal-consequence calculations: exactly the code that needs exhaustive boundary tests (the day before / day of / day after each window; leap-year and timezone edges; the bypass-substitution matrix). A wrong result tells a pilot they are legal to fly when they are not. **Recommend a table-driven unit suite per rule plus golden fixtures, gated like the captadel eval suite.**

3.2 CRM — security-rule and conversion coverage

`spec-crm.md` describes `assertAdmin(req)` enforced on every callable, a proposed narrower `crm` claim, and lead→account conversion (`grantSchoolLicence`) — none with tests. Mirror the instructor-dashboard approach: **Firestore-emulator rule tests** for every callable (admin / non-admin / proposed `crm` claim) and a conversion-flow test.

3.3 Freshness pipeline — validation coverage

Ingests/diffs the public corpus and rebuilds indexes. No tests for diff correctness, index-rebuild integrity, or the data-boundary guarantee ("never touches personal data"). Recommend: diff-correctness fixtures (known-input → expected changelog), an index-rebuild smoke test, and a boundary assertion that no PII crosses into deployed artifacts.

4. Cross-cutting coverage improvements

- **Arabic parity is under-tested.** Plan target is AR pass-rate within 5% of EN across the *whole* suite, but AR cases are sprinkled and lag. Make AR parity a measured CI gate, not a per-feature afterthought.
- **Multi-turn conversation cases** are deferred to a later wave; only Tier-4 (5 cases) in the sales-agent suite exercises realistic dialogue. RAG-side multi-turn (context carry, follow-up disambiguation) is essentially uncovered today.
- **CI enforcement exists on paper** (`refusal-protocol` §7.4: zero-tolerance on refusal/injection regression, AR-within-5% bar). Confirm `.github/workflows/eval.yml` actually implements these gates in the code repo — the spec is ahead of verified reality.
- **No reconciliation between the two suites.** `test-ready.md` (55 cases, "≥38") and the captadel plan (≥95) use different baselines and runners. A single coverage dashboard across both would prevent each from being read as "done" in isolation.

5. Recommended priority order

1. Add the four §7.2 refusal fixtures (already written) — fastest win, safety-critical.
 2. Define and build the **currency-tracker** boundary unit suite — highest real-world risk.
 3. Add **tool-use eval cases** before any compute tool ships.
 4. Add **CRM + freshness-pipeline** emulator/validation tests (currently zero).
 5. Promote **AR parity** and **multi-tenant isolation** to measured CI gates.
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Generated as a spec-vs-test gap audit; no executable coverage tooling applies to this repository. Update alongside the implementation repos as suites land.